CHAPTER L-01

WHAT LIFE INSURANCE INVOLVES

Chapter Introduction

We have seen some aspects related to Insurance in the common chapters. However, when it comes to Life insurance, we need to look at them more deeply.

- ✓ An asset
- ✓ The risk insured against
- ✓ The principle of pooling
- ✓ The contract

Let us now examine the features of life insurance. This chapter will take a brief look at the various components of life insurance mentioned above.

Learning Outcomes

A. Life insurance business –Components, human life value, mutuality
B. Risks and Life Insurance

A. Life insurance business – Components, human life value, mutuality

a) The Asset – Human Life Value (HLV)

We have already seen that an asset is a kind of property that yields value or a return. For most kinds of property both the value and loss of value amounts can be measured in precise monetary terms.

Example

If the estimated damage of a car meeting an accident is Rs 50000, the insurer will compensate the owner for this loss.

How do we estimate the amount of loss when a person dies?

Is he worth Rs. 50,000 or Rs. 5,00,000?

An Agent must be able to answer the above question when meeting a customer. Based on this the agent can determine how much insurance to recommend to the customer. It is in fact the first lesson a life insurance agent must learn.

Luckily we have a measure, developed almost seventy years ago by Prof. Hubener. It is known as **Human Life Value (HLV)** and is used worldwide.

The HLV concept considers human life as a kind of property or asset that earns an income. It thus measures the value of human life based on an individual's expected net future earnings. Net earnings means the income a person expects to earn each year in the future, less the amount he would spend on himself. It thus indicates the economic loss a family would suffer if the wage earner were to die prematurely. These earnings are capitalised, using an appropriate interest rate to discount them.

Although there are multiple parameters used to calculate HLV including taking into account inflation, wage rise, future earning capacity etc., a simple thumb rule to calculate HLV is to determine the amount that would generate the annual income the family would be needing by way of interest. In other words HLV is the annual contribution for the family by the breadwinner divided by the prevailing rate of interest.

Example

Mr. Rajan earns Rs. 1,20,000 a year and spends Rs. 24,000 on himself. The net earnings his family would lose, were he to die prematurely, would be Rs. 96,000 per year. Suppose the rate of interest is 8% (expressed as 0.08).

Human-Life-Value (HLV) = Annual Contribution for Dependents ÷ Rate of Interest

HLV = 96000/ 0.08 = Rs. 12,00,000

HLV helps to determine how much insurance one should have for full protection. It also tells us the upper limit beyond which providing life insurance may not be reasonable.

In general, the amount of insurance should be around 10 to 15 times one's annual income. Thus one should grow suspicious if Mr. Rajan was to ask insurance of Rs. 2 crores, while earning only Rs. 1.2 lakhs a year. The actual amount of insurance purchased would depend on factors like how much insurance one can afford and would like to buy.

B. Risk and Life Insurance

As we have seen above, life insurance provides protection against those risk events that can destroy or reduce the value of human life as an asset. There are three kinds of situations where such loss can occur. They are typical concerns which ordinary people face.

Diagram 1: Typical concerns faced by ordinary people

General insurance on the other hand typically deals with risks that affect property – like fire, loss of cargo while at sea, theft and burglary and motor accidents. They also cover events leading to loss of name and goodwill. These are covered by liability insurance.

Finally there are risks that can affect the person. Termed as personal risks, these may also be covered by general insurance.

Example

Accident insurance which protects against losses suffered due to an accident.

a) How exactly does life insurance differ from general insurance?

General Insurance	Life Insurance
• Indemnity: General insurance policies, with the exception of Personal Accident Insurance, are usually contracts of indemnity i.e. after an event like fire, the insurer assesses the exact amount of loss that has occurred and compensates only that amount of loss – no more, no less.	• **Assurance:** Life insurance policies are contracts of assurance. • The amount of benefit to be paid in the event of death is fixed at the beginning of the contract. • An assured sum is paid to the nominees or beneficiaries of the insured when he dies.
• Duration: The contract is generally short period or for one year renewable basis	• The contract is generally long term though some one year renewable contracts are also prevalent
• Uncertainty: In general insurance contracts, the concerned risk event is	• There is no such question Death is certain once a person is born. What is

uncertain. No one can be certain about whether a house would catch fire or a car meet an accident.	uncertain is the time of death. Life insurance offers protection against the risk of premature death.
• Increase in probability: In case of General insurance perils like fire or earthquake, the probability of happening of the event does not increase with time.	• In life insurance the probability of death increases with age.

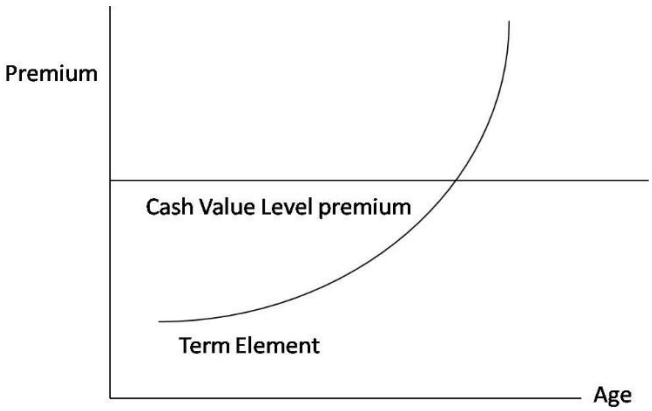
b) Nature of life insurance risk

Since probability of death increases with age, lower premiums are charged for those who are young and higher premiums for older people. One result was that old individuals who were in good health, tended to withdraw while unhealthy members remained in the scheme. Insurance companies faced serious problems as a result. Their attempts to develop life insurance policies that people could afford led to the development of level premiums.

c) Level premiums

The level premium is fixed such that it does not increase with age but remains constant throughout the contract period. This means premiums collected in early years is more than the amount needed to cover death claims of those dying when young, while premiums collected in later years are less than what is needed to meet claims of those dying at higher ages. The level premium is an average of both. The excess premiums of earlier ages compensate for the deficit of premiums in later ages. The level premium feature is illustrated below.

Diagram 2: Level Premium



Level premiums are required because life insurance contracts are long term insurance contracts that run for 10, 20 or many more years. The concept of level premiums, do not arise for general insurance policies, which are typically short term and expire annually.

Example

The level premium rate is arrived at by the insurers based on the mortality (probability of death) during the term of the policy as the age of the insured would increase every year. The rate once decided shall be constant for the entire term of the policy.

d) The Principle of Risk Pooling and Life Insurance

We have already discussed the Principle of Pooling and Mutuality earlier. The pooling principle plays two specific roles in life insurance.

i. It **provides protection against the economic loss arising as a result of one's untimely death**. This is done by creating a fund that pools the contributions of many who have purchased a life insurance contract.

e) The Life Insurance Contract

The Policy document is the **evidence of the insurance contract** which a details all the terms and conditions of the **insurance**.

The contract states the sum assured of the life insurance policy. Life insurance is regarded a **financial security** as the sum Insured is guaranteed by the contract. The guarantee implies that life insurance is managed efficiently and conservatively; strongly regulated and strictly supervised.

Since Life insurance contracts involve both risk cover and savings, they are often compared with financial products. They are also seen as a way of holding wealth than as protection. Indeed, many life insurance products have a large cash value or savings component which can form a significant part of an individual's savings. Some do argue that it may be better to buy only Term Insurance from an insurance company and invest the balance premiums in instruments that yield higher returns.

Let us consider the arguments for and against traditional cash value insurance contracts.

a) Advantages

i. Insurance has historically been proven as a **safe and secure investment offering** a minimum guaranteed rate of return, which may increase with contract duration.

ii. Regularity of premium payments requires compulsory planning of one's savings and results in savings **discipline**.

iii. The Insurer takes care of professional investment management and **frees** the **individual** of this responsibility

iv. Insurance **provides liquidity**. The insured can take a loan on or surrender the policy and convert it into cash.

v. Both cash value type life insurance and annuities may enjoy some **income tax advantages.**

vi. Insurance may be **safe from creditors' claims**, generally in the event of the insured's bankruptcy or death.

b) Disadvantages

i. As insurance gives relatively fixed and stable returns, it can be seriously affected by inflation.

ii. High marketing and other initial costs reduces the amount of cash value accumulated in earlier years of life insurance policies.

iii. The guaranteed yield may be below that of other financial instruments

Test Yourself 1

How does diversification reduce risks in financial markets?

I. Collecting funds from multiple sources and investing them in one place

II. Investing funds across various asset classes

III. Maintaining time difference between investments

IV. Investing in safe assets

Summary

a) Asset is a kind of property that yields value or a return.

b) The HLV concept considers human life as a kind of property or asset that earns an income. It thus measures the value of human life based on an individual's expected net future earnings.

c) The level premium is a premium fixed such that it does not increase with age but remains constant throughout the contract period.

d) Mutuality is one of the important ways to reduce risk in financial markets, the other being diversification.

e) The element of guarantee in a life insurance contract implies that life insurance is subject to stringent regulation and strict supervision.

Key Terms

1. Asset
2. Human Life Value
3. Level premium
4. Mutuality
5. Diversification

Answers to Test Yourself

Answer 1 - The correct answer is II.